# Standard Operating Procedure

**Bioethanol Distillation Column — Unit D-201**

| | | | |
|---|---|---|---|
| **Document ID** | SOP-D201-014 | **Revision** | Rev 4 |
| **Effective date** | 2026-05-01 | **Review due** | 2028-05-01 |
| **Owner** | Process Engineering | **Classification** | Internal / Controlled |

## 1. Purpose & Scope

This procedure defines the safe and normal operation of distillation column D-201, which concentrates fermentation beer (approx. 8–12% v/v ethanol) to a rectified ethanol distillate of nominally 95% v/v. It applies to all operators and shift engineers on the bioethanol plant. This is a controlled operating document; it is not a substitute for the site permit-to-work system.

## 2. Process Description

Fermented beer is preheated and fed to the column. Overheads are condensed; part of the condensate is returned as reflux and the balance drawn off as rectified ethanol distillate. Bottoms (stillage) are drawn from the reboiler sump. Column heat is supplied by low-pressure steam to the reboiler E-201.

## 3. Normal Operating Parameters

| Parameter | Normal | Safe range | Units |
|---|---|---|---|
| Feed flow rate | 12.0 | 9.0 – 14.0 | $m^3/h$ |
| Feed temperature | 85 | 80 – 92 | °C |
| Column top pressure | 1.15 | 1.05 – 1.30 (max 1.50) | bar a |
| Reflux ratio (L/D) | 3.0 | 2.5 – 3.8 | – |
| Reboiler steam flow | 2.4 | 1.8 – 3.0 (max 3.3) | t/h |
| Top (overhead) temperature | 78.3 | 77.5 – 79.5 | °C |
| Bottoms temperature | 103 | 98 – 107 | °C |
| Distillate purity (target) | 95.0 | ≥ 94.0 | % v/v |

*Operating outside the stated safe range requires shift engineer authorisation. Exceeding a listed maximum is prohibited without an approved deviation.*

## 4. Startup Procedure (summary)

4.1 Confirm permit-to-work status and that E-201 steam isolation is available.
4.2 Verify instrument air, cooling water to condenser C-201, and level in reflux drum.
4.3 Establish total reflux; admit steam slowly, raising reboiler duty in steps of ~0.3 t/h.
4.4 When top temperature stabilises near 78 °C, begin feed at 9 $m^3/h$ and increase to normal.
4.5 Set reflux ratio to 3.0; begin distillate draw when purity ≥ 94% v/v is confirmed by analyser.

## 5. Normal Shutdown (summary)

Reduce feed to minimum, stop distillate draw, cut reboiler steam gradually, maintain reflux and cooling until the column cools below 60 °C, then isolate.

## 6. Alarms & Trips

| Tag | Description | Alarm | Trip / action |
|---|---|---|---|
| PT-201 | Column top pressure | 1.35 bar a (high) | 1.50 bar a → auto steam cut |
| TT-205 | Bottoms temperature | 108 °C (high) | 112 °C → auto steam cut |
| LT-203 | Reflux drum level | 15% (low) / 85% (high) | 8% low → distillate pump trip |
| AT-207 | Distillate purity | 93.5% v/v (low) | Divert to slops on sustained low |

*Trips are safety functions. Overriding, bypassing or resetting a trip requires a qualified engineer and an approved override permit. Operators must not defeat a trip to keep the unit running.*

## 7. Safety Precautions

Ethanol and its vapours are highly flammable (see the Ethanol Safety Data Sheet). No hot work, welding or open flame is permitted on or near D-201 without a valid hot-work permit and gas testing. Required PPE for sampling and field work: safety glasses/goggles, chemical-resistant gloves, flame-retardant coveralls, and safety footwear. Ensure bonding/earthing when transferring ethanol. Report any liquid or vapour leak immediately.

## 8. Troubleshooting (quick guide)

| Symptom | Likely cause | First checks (within SOP limits) |
|---|---|---|
| Distillate purity low | Reflux too low / feed upset | Raise reflux ratio toward 3.8; verify feed rate & temp; check analyser AT-207 |
| High energy use per litre | Reflux higher than needed | Trim reflux toward 2.5 while holding purity ≥ 94%; confirm feed preheat |
| Flooding / high dP | Vapour load too high | Reduce reboiler steam within range; reduce feed |
| Top temperature rising | Loss of reflux / pressure rise | Check reflux pump & drum level; watch PT-201 |

## 9. Authorisation & Human Sign-off

The following always require a qualified/shift engineer and, where noted, a permit: operating any parameter outside its safe range; changing or bypassing an alarm/trip setpoint; isolation, breaking containment, or hot work; startup after a trip. Automated systems and assistants may advise but must not authorise these actions.